Version 1.0; Publication date: 2026-11-01
COSI — Glossary

VDW
Die Wellpappenindustrie



COSI 1.0
Corrugated Standard Interface

Open industry standard for machine data exchange
in corrugated cardboard manufacturing

Imprint

COSI — Glossary

COSI — Corrugated Standard Interface

Version	1.0 (prepublication)
Updated	2026-10-16
Publication date	2026-11-01
Status	Prepublication — package approved
Scope	Standard definitions for all COSI domain terminology

Publisher / working group

VDW (Association of the German Corrugated Cardboard Industry e.V.)

Authors

Michael Sachs

Robert Schmid

Daniel Malolepszy

Dennis Strasser

Tobias Winkler

Technical editors

Technical specification development and quality assurance.

André Seifert (ePS)

Robin Boeckxstaens (OMP)

Contact

Christiane Trumpp

transparente-daten@vdw-da.de

Disclaimer

The PDF edition is a reproduction of the Markdown source published in the official COSI GitHub repository. In the event of any discrepancy, the Markdown source for the corresponding document version shall prevail.

https://github.com/vdw-da/COSI-Corrugated-Standard-Interface

Impressum / Legal Notice

Publisher: **Verband der Wellpappen-Industrie e.V. (VDW)**, a registered association. Registered address: Markgrafenstraße 19, 10969 Berlin, Germany. Represented by the Chair of the Board, Dr. Steffen P. Würth. Association register: VR 40431 B, Amtsgericht Charlottenburg (Charlottenburg Local Court). VAT identification number: DE 111 607 830. Telephone: +49 30 7543880-0. Email: info@wellpappen-industrie.de.

These publisher details are drawn from VDW's official Impressum — Legal Notice. For COSI questions and suggestions: transparente-daten@vdw-da.de. The repository's Apache License 2.0 governs COSI licensing; website-specific copyright restrictions are not imported into this repository.

Privacy

VDW's full privacy policy: Datenschutzerklärung — Privacy Policy.

Use of GitHub and sharing of content

We use the GitHub platform to provide, manage and collaborate on digital content. For this purpose, selected repositories, files, documentation and project-related information within our GitHub organisation may be made available to external users. Personal data (such as GitHub usernames and email addresses) may be processed where they are included in the shared content, arise through use of the platform or are needed to grant access. Processing serves to provide and share the relevant content and to collaborate with external users.

The platform provider is GitHub, Inc., USA. For details of how it handles your personal data, please see the GitHub Privacy Statement.

The company is certified under the EU–US Data Privacy Framework (DPF). The DPF is an arrangement between the European Union and the United States intended to ensure compliance with European data protection standards when data are processed in the United States. Each DPF-certified company undertakes to comply with these standards. Further information is available in the provider's DPF entry.

Inquiries by email, telephone or fax

If you contact us by email, telephone or fax, we store and process your inquiry, including all personal data it contains (name, inquiry), to handle your request. We do not disclose these data without your consent.

If your inquiry concerns documents or content provided through GitHub, it may be forwarded to the responsible working groups where this is necessary to assess or further process the inquiry. Inquiries that are not pursued further are deleted.

These data are processed on the basis of Article 6(1)(b) GDPR where your inquiry relates to the performance of a contract or is necessary to take steps prior to entering into a contract. In all other cases, processing is based on our legitimate interest in effectively handling inquiries addressed to us (Article 6(1)(f) GDPR), or on your consent (Article 6(1)(a) GDPR) where requested; consent may be withdrawn at any time. We retain data you send us through contact inquiries until you ask us to delete them, withdraw your consent to their storage, or the purpose for storing them no longer applies (for example, once your request has been dealt with). Mandatory statutory provisions, particularly statutory retention periods, remain unaffected.

Term	Abbreviation	Standard definition	Notes / usage
Corrugator	—	Complete corrugating line from roll handling/infeed through wet end and dry end to stacker/outfeed.	Use when referring to the full corrugated board production line.
Converting machine	—	Any downstream processing machine (flexo folder gluer, die-cutter, casemaker, printer, laminator, etc.) from infeed/prefeeder to outfeed/palletiser.	Use "converting" consistently instead of mixed German/English terms.
MQTT	—	Message Queuing Telemetry Transport — publish-subscribe transport for COSI IT/OT crossings.	Horizontal exchanges within IT or OT may use other protocols. TLS :8883; QoS 1 recommended; dedicated Will QoS 1 required.
Quality of Service	QoS	MQTT delivery guarantee level: 0 (at most once), 1 (at least once), 2 (exactly once).	QoS 1 is the baseline for production environments.
Heartbeat message	HBT	Periodic message at a defined cycle (typically 1–5 s) independent of state changes.	Used for continuous performance values.
Change-of-status message	COS	Event message emitted whenever a machine or system state changes.	Used for discrete events (run/stop/fault, stack discharge, setup triggers).
Asset class	`assetClass`	Business domain of a message — determines which payload schema applies.	Values: `performance`, `setup-corrugator`, `setup-converting`, `setup-response`, `maintenance-document`, `maintenance-measurement`, `quality`, `lastwill`. See cosi-message-properties-v1.0.pdf.
Message ID	`messageID`	Globally unique UUID v5 assigned to one message instance.	Distinct instances need distinct generation inputs even for identical content. Retries/replays preserve the assigned ID across restart.
Correlation ID	`correlationID`	UUID v5 linking related messages; required for `setup-response` and equal to the original setup messageID.	Optional for other asset classes.
UUID v5	—	Name-based UUID using namespace and name; identical inputs produce the same UUID.	Lowercase hyphenated form for messageID, correlationID and connectionId. Producers allocate distinct inputs; v5 alone neither guarantees uniqueness nor performs de-duplication.
One-time data	—	Static or rarely changing reference data.	Documents and task definitions use o `netime`; COSI v1.0 does not define an automatic document-update service.
Machine boundary	—	The complete machine within agreed process limits (corrugator: full line; converting: infeed -> outfeed).	All input/output/consumption values refer to this boundary.
Machine state	—	Discrete operating mode of the machine.	Use only: `unknown`, `stop`, `run`, `fault`, `setup`, `maintenance`, `offline`.
Run	`run`	Production running at any line speed >0 within defined boundary.	Standard state label for performance reporting.
Stop	`stop`	Machine not producing; line speed = 0 within boundary.	Planned and unplanned stops are sub-categories.
Fault	`fault`	Stop caused by a failure requiring intervention; may be accompanied by failure code.	Use with ISO-aligned failure mode codes.
Setup state	`setup`	Machine operating state during setup activities.	Distinct from setup instructions for a planned execution.

Term	Abbreviation	Standard definition	Notes / usage
Setup execution	—	Instructions for one planned production execution: possibly several corrugator orders, or a converting processing operation.	Pending revisions and continuation after interruption preserve identity; a separately scheduled execution has a new identity. Corrugator setupId is unique within site/machine.
Maintenance	`maintenance`	Machine is not producing due to maintenance activities.	
Offline	`offline`	Machine intentionally taken out of operation.	Final message before a controlled shutdown (see Technical Guide Section 2.5.1).
Changeover	—	Planned activity between jobs, from last good sheet of previous job to first good sheet of next job.	Often called "Setup" in colloquial usage; *Setup time* KPI uses this definition.
Setup time	—	Duration measured from the last good sheet of the previous order to the first good sheet of the current order — includes tool changes, machine adjustments, test runs, and material changeovers.	Primary setup KPI. See Technical Guide Section 4.4.
Stopper	—	Any event causing speed reduction or stop within the boundary.	Prefer "stopper" over mixed terms such as disturbance, blocking, etc.
Throughput	—	Output per time.	Use units `m²/h` or `pcs/h`; specify which.
OEE	—	Overall Equipment Effectiveness combining availability, performance, and quality.	If calculated, document formula and boundary.
Mean Time Between Failures	MTBF	Average operating time between successive failures of a component or system.	Performance and maintenance KPI.
Mean Time To Repair	MTTR	Average time to restore a failed component or system to operation.	Maintenance KPI; lower values indicate faster recovery.
Job reference	—	Unique identifier for an order/job (from cut list or job list).	Stable across systems (ERP/MES/machine).
Article number	—	Unique identifier of a product variant or SKU.	May repeat across jobs.
FEFCO	—	European Federation of Corrugated Board Manufacturers — publisher of the standardised box-style code system.	The code identifies the box construction (e.g., 0201 = standard slotted carton). Used as `boxType` in converting setup.
FSC	—	Forest Stewardship Council certification — verifies responsibly sourced paper fibres.	Levels: 100%, 75%, Recycled. Used as `requiredFSC` in corrugator setup.
Target speed	—	Reference speed supplied by ERP/MES according to the plant's business rules.	Derivation may vary by implementation; COSI prescribes no baseline calculation method.
Running length	lm	Linear metres of board processed within boundary.	Use SI metre; `lm` as field label is acceptable.
Area	m²	Board area at the defined counting point or location.	Input/output counters are cumulative within scope/period; logistics quantities are current contents. Article changes alone do not reset counters.
Weight	t	Metric tonnes of board processed within boundary.	
Grammage	gsm	Mass of paper per unit area.	Expressed in `g/m²`; use the schema field name `gsm`.
Pieces	pcs	Discrete material or product units at an identified counting point or location.	Define a consistent counting basis within the scope/period; compare only compatible quantities.

Term	Abbreviation	Standard definition	Notes / usage
Stacks	`stacks`	Number of physical stacks.	Performance input/output counts are cumulative; logistics counts are current contents. Not sheets per stack.
Stack count	`stackCount`	Requested number of sheets per stack in corrugator setup.	Three stacks of 200 sheets have stackCount 200, stacks 3 and pieces 600.
Stack height	`stackHeight`	Physical stack height in millimetres.	Setup is a requested setting; quality checks evaluate physical height.
Counting period	—	Period within which cumulative readings of the same scope are comparable.	Stops do not reset it. Reset/scope change must be distinguishable; do not calculate deltas across periods. Incomplete totals are not zero.
Allocated consumption	—	Calculated share of a measured total for a stated source period and allocation basis.	Shares plus identifiable remainder reconcile to the source total; measured-total-only reporting is permitted.
Current contents	—	Material present at a location at observation time.	Logistics quantities are inventory observations, not accumulated throughput.
Left-to-right	L2R	The order in which orders are physically arranged across the corrugator web, numbered from left to right when looking downstream (towards stacker/in the direction of board travel). Position 0 is the leftmost order on the web.	Used as `l2rSequenceNumber` on each order in the setup-corrugator payload to define the layout.
Flute	—	Wave-shaped corrugated paper layer between liners; defines board profile and stacking strength.	Standard flute types: A, B, C, E, F, BC.
Caliper	—	Thickness of paper or board, measured in mm.	Used as both paper specification and quality measurement.
Trigger	—	Timestamped setup- or process-relevant event (e.g., tool installed, wash program started).	See the COS definition and the applicable payload schema for message usage.
Periphery	—	Auxiliary devices connected to the machine (e.g., glue unit, ink kitchen, stacker robot).	The applicable setup schema defines where periphery settings are represented.
Single-facer	SF	Corrugator section that bonds one fluting layer to one liner.	A corrugator typically has one to three single-facers.
Double-backer	DB	Corrugator section that bonds the final liner to assembled board, completing the corrugated structure.	Final wet-end section before cutting and stacking.
Fanfold	—	Periphery device producing accordion-folded corrugated material instead of cut sheets.	Order-assignable.
Tape applicator	—	Periphery device applying reinforcement or splice tape during corrugation.	
Flexographic printing	flexo-print	Direct relief-printing process using flexible plates on cylindrical print stations.	Standard print process for corrugated board.
Printing plate	cliché, plate	Flexible plate mounted on a flexo-print cylinder, carrying the artwork for one ink colour.	"Cliché" is the European term; standard field name is `plateId`.
Die-cutter	—	Converting periphery that uses a shaped tool (die) to cut and crease blanks from board sheets.	Rotary or flat-bed variants.
Folder-gluer	FFG	Converting machine that folds and glues die-cut blanks into finished boxes.	Often combined with flexo printing into a Flexo Folder Gluer (FFG) line.

Term	Abbreviation	Standard definition	Notes / usage
Stitcher	—	Converting periphery that joins box panels with metal staples instead of glue.	Used for large or heavy-duty boxes.
Palletizer	—	Converting periphery that stacks finished boxes or bundles onto pallets in defined patterns.	Includes strapping, wrapping, and label placement.
Failure mode	—	Standardized classification of a fault, adapted from ISO 14224:2016, Annex B.	`faultCode` — closed set of 40 three-letter codes (cosi-corrugated-faults-v1.0.pdf), plus optional vendor `internalCode`.
Failure severity	—	Impact level of a fault: `info`, `warning`, `critical`.	Use these exact lowercase values.
Quality check	—	Atomic check with counted outcome in a `checks` array under its label, limits and unit.	Numeric or qualitative OK/NOK; receiving applications build reports. A message is not a complete/cumulative report.
Check result	—	`OK` or `NOK` against the applicable numeric or qualitative criterion.	Qualitative limits may be null; omit exception unless a numeric bound was violated. Missing checks do not establish OK.
First-good-sheet	—	First sheet meeting all quality criteria after a setup or order change.	Marks the end of setup time and the start of productive run.
Warp	—	Deviation of a board sheet from flat, expressed as a percentage.	Quality KPI; high warp causes feeding and stacking issues.
Register	—	Alignment between successive print stations, or between print and die-cutting.	Reported as longitudinal offset in mm; high register error indicates print mis-alignment.
Fishtailing	—	Defect where sheet edges fan out unevenly during folding, giving a fishtail shape.	Quality KPI in folding/gluing; reported as percentage of affected sheets.
Maintenance document	`maintenance-document`	One-time message carrying static reference data for a machine component: manual links and service task definitions with frequency triggers.	`assetClass=maintenance-document`, `messageType=onetime`. COSI v1.0 does not define an automatic document-update service.
Maintenance measurement	`maintenance-measurement`	Event-driven observation at its original timestamp, with required failure and document-reference objects whose individual fields may be null as permitted.	`assetClass=maintenance-measurement`, `messageType=cos`. Triggers when a threshold is crossed or a condition is detected.
Service task	—	A defined maintenance activity for a component, with a frequency trigger (runtime hours, area, pieces, or cycles) and a linked procedure document.	Frequency source must be specified (e.g., motor runtime, production counter).
Frequency trigger	—	The counter or threshold that initiates a service task: `h` (hours), `m²` (area), `pcs` (pieces), or `cycles`.	Defined in the maintenance-document payload.
Operational technology	OT	Hardware and software directly monitoring or controlling physical production — Levels 0–3 of the architecture (sensors, PLCs, machine control systems, interpreters).	See Technical Guide Section 2.4.
Information technology	IT	Business and data systems above the production floor — Levels 4–5 (local data servers, ERP/MES, central/cloud services).	See Technical Guide Section 2.4.

Term	Abbreviation	Standard definition	Notes / usage
North-south	—	Communication crossing the IT/OT boundary. Permitted only through the Level 3.5 message broker; direct crossings are prohibited in both directions.	Setup, response and machine observation messages are brokered north-south flows. See ADR-000010.
East-west	—	Peer-to-peer communication inside one zone (controller <-> controller in OT, ERP <-> MES in IT). MQTT is optional within a zone; direct hardware-to-hardware control is out of scope of this standard.	See Technical Guide Section 2.4 and ADR-000010.
Schema identifier	`schemaId`	Required full payload schema identifier, including version, matching assetClass.	Example: `cosi:performance-payload:1.0`; topic has no version segment.
Connection identifier	`connectionId`	Globally unique UUID v5 for one MQTT connection using Last Will.	Shared by its Will and normal equipment messages; fresh on reconnect/restart, distinct from MQTT client ID and messageID.
Last Will	`lastwill`	Optional OT connection-loss notification registered with MQTT.	Dedicated message, null timestamp and empty payload; QoS 1, non-retained, Keep Alive enabled. Consumer analysis/handling is outside COSI.

Imprint and Privacy Notice (Impressum / Datenschutz)